

Christopher Kuizon

I'm a machine learning engineer. I spend most of my time on the part after training: checking whether a model still works on messy, real inputs.

ML Engineer at TrueCar. CS and Business at USC, continuing into an M.S. in CS. Based in Los Angeles.

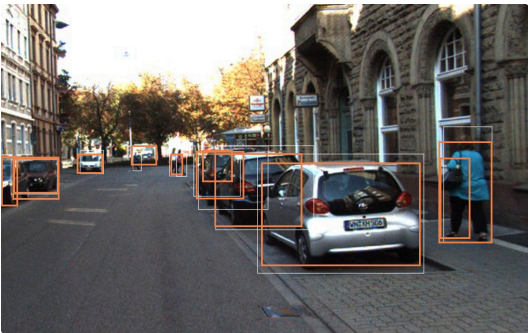
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GitHub github.com/Khriis-K

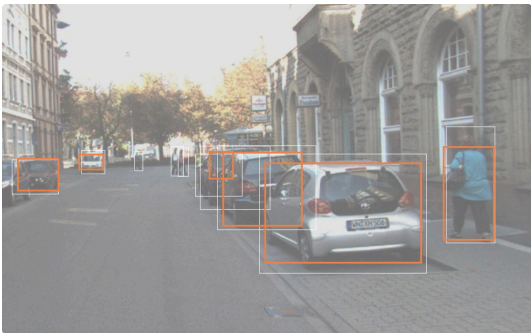
INTERACTIVE VERSION

khriis-k.github.io

Live demos, video, and the fog slider



clear · 12 detections ≥ 0.25



fog severity 0.60 · 6 detections ≥ 0.25

Same frame, same detector, more fog. Orange boxes are detections; thin white outlines are the labeled objects. Fog and detector from the Edge Perception Reliability Lab. On the website this is a slider. Frame: KITTI (Geiger et al., CVPR 2012), cropped, CC BY-NC-SA 3.0.

SELECTED WORK

Edge Perception Reliability Lab

Personal project, Sep 2026 to now

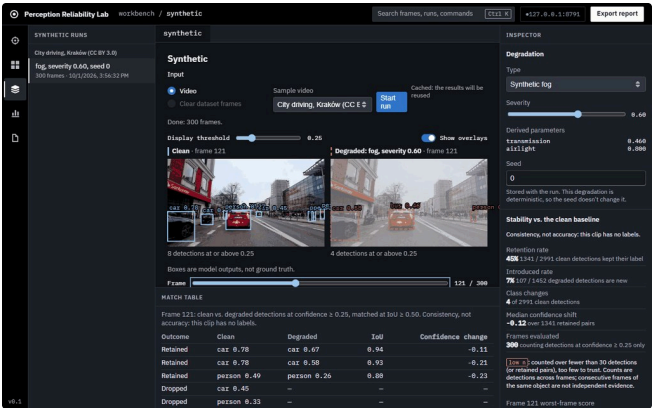
Does a detector trained on clear roads still work in fog, snow, and at night? This lab runs YOLOX-Nano over 2,000 driving frames in 8 weather and lighting conditions, and compares real bad weather against simulated degradation.

On a 300-frame benchmark with 2,946 labeled objects, mAP dropped 85% under severe simulated fog. The scoring engine that classifies hits, misses, false alarms, and class mix-ups has 800+ automated tests.

Python, FastAPI, React, TypeScript

Project khriis-k.github.io/Edge-Perception-Reliability-Lab

Source github.com/Khriis-K/Edge-Perception-Reliability-Lab



The workbench during a synthetic fog run on 300 frames of city driving in Kraków.

MyCouncil

Personal project, Nov 2025 to now

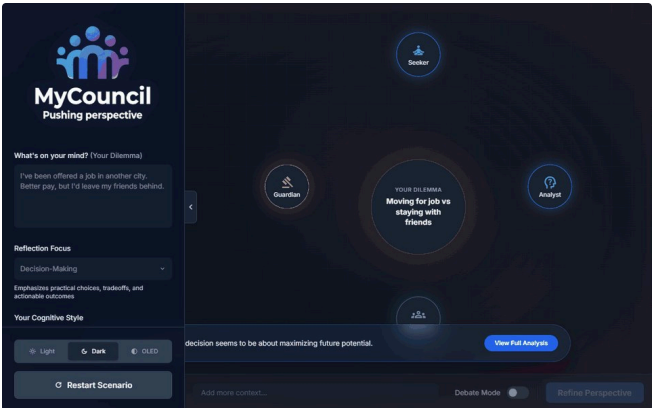
You bring a hard decision and 16 AI counselors with different thinking styles argue it out. You can debate the dilemma, or talk to one counselor alone.

The hard part was memory. Counselors recall what you said in earlier sessions through a retrieval pipeline with BGE embeddings and a reranker. It improved relevant-context recall by 168% on 160 held-out tests, measured within a 240-case suite that also tracks ranking quality and latency.

Python, Node.js, React, TypeScript

Project khriis-k.github.io/MyCouncil

Source github.com/Khriis-K/MyCouncil



A council forming around "move for the job, or stay with friends?"

What caused 20 years of runway incursions

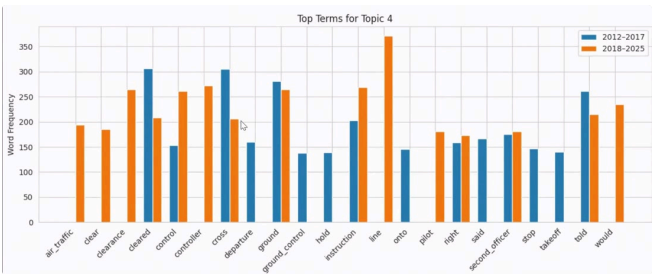
USC Aviation Safety & Security Program, Mar to Jun 2025. Team of eight; I built the topic models.

NASA's Aviation Safety Reporting System holds pilot and controller narratives of runway incursions going back to 2001. Researchers have mostly coded them by hand. We used NLP to find the recurring causes automatically and to see how they changed over time.

I built the LDA and BERTopic pipelines, including expanding aviation abbreviations in the preprocessing step and tuning UMAP and HDBSCAN so the topics separated cleanly. Then I compared topics before and after 2017. We presented the findings to 50+ program representatives.

Python, gensim, BERTopic, sentence-transformers, React, FastAPI

Project [khriis-k.github.io/ASRS-Dashboard](https://github.com/khriis-k/ASRS-Dashboard)
Source github.com/Khriis-K/ASRS-Dashboard



One topic, two periods. "Line" and "clearance" appear only after 2018, while "hold" and "takeoff" drop out.

Rebuilding sales attribution at TrueCar

Machine Learning Engineer, May 2026 to now. Proprietary, so no code here.

TrueCar's legacy sales attribution algorithm used a probabilistic ranking built on a naive Bayes model.

I rebuilt the ranking and trained an XGBoost model on 52 million labeled pairs with 10 target-encoded features.

XGBoost, Spark, AWS

Change versus the legacy system, at over 95% confidence

Billable sales in production	+124%
Monthly match volume	+196%
Match accuracy	+19%
Top-1 accuracy vs. naive Bayes	+59%
AUC-ROC	+30%
Mean reciprocal rank	+29%

TrojanSafe Housing

A web app that maps historical crime data from USC Department of Public Safety daily incident logs, so students and their families can compare off-campus housing. Private repo, code available on request.

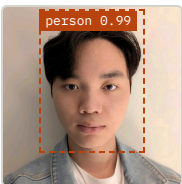
Counsel Reflect

A tool for reviewing counseling conversations turn by turn, with 81 emotion, safety, and factuality metrics. I helped design the interface and React dashboard, which scored 78.8 and 75.2 on usability against a 68 benchmark.
counselreflect.com

EXPERIENCE

Machine Learning Engineer, TrueCar	May 2026 to now
Full-Stack Software Engineer, Counsel Reflect	May 2025 to Aug 2026
Fine-tuned RoBERTa classifiers to 80 to 92% accuracy across three empathy dimensions.	
Data Analyst, USC Aviation Safety & Security Program	Mar to Jun 2025
Teaching Assistant, USC machine learning courses	Jan 2025 to now
Four courses, from intro AI to NLP and generative AI. I've mentored 100+ students across 90+ projects.	
Social Chair, EscapeSC	Fall 2026
USC's escape room club. We design and build rooms on campus.	
Co-Chair, Viterbi Undergraduate Student Association	Feb 2024 to Aug 2026
Represented 50+ engineering clubs to the dean and started the school's first AI ethics panel.	

ABOUT



I'm most interested in ranking, retrieval, and applied ML that has to hold up in the real world, whether it's serving millions of users or running somewhere mistakes are costly.

I also TA machine learning courses, so if I'm not building something, I'm probably explaining why someone else's model isn't training. Otherwise I'm with EscapeSC, USC's escape room club, where we build rooms on campus. My friends and I are 17 rooms in.

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